Adarsh Dwivedi

Software Engineer

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Final-year CS engineer who ships, deploys and tests. Built an agentic platform on Node.js and MCP covered by 719 tests in CI, an analytics platform joining 59,985 synthetic police records into one queryable graph and deployed for a state force in 50 days, and a caching proxy on PyPI that fails open rather than failing the request. At Deloitte, migrated a production agent onto a self-hosted server without changing its public API, reversible through a single setting.

EXPERIENCE

Deloitte India, Digital Excellence Centre **Jun 2026 - Aug 2026**
AI Engineer (Agentic AI) Intern Gurugram, India

- Replaced request-time graph assembly with a graph factory that registers one static graph and carries per-request variation in run state, verifying the migrated output identical to the original field by field behind an unchanged API.
- Held the public API fixed so the front end and test collections needed no change, and made the migration reversible through one setting, EXECUTION_BACKEND, keeping the original execution path byte for byte so rollback needs no redeploy.
- Docker was unavailable, so worked out and documented a Docker-free deployment of FastAPI, Aegra and Playwright MCP on a shared Linux server against PostgreSQL, after tracing unrelated-looking failures to an unsupported runtime and rebuilding on Python 3.13.

PROJECTS

AGENTIQ, Agentic API Testing and Security Validation *(B.Tech final-year project)* | **Live** | **GitHub** Node.js, Express, MCP, Docker, Vitest

- Built an API testing and security platform on Node.js and MCP that turns a URL and plain-English intent into executable test cases with status, JSONPath, header and body assertions, grounded in an imported OpenAPI 3.1 document.
- Designed the tool layer as 21 Zod-validated tools across 6 risk classes with per-host grants, a filesystem jail and a process sandbox, and isolated state-mutating endpoints so a test run cannot corrupt the target.
- Covered the system with 719 tests in CI and tied test-generation confidence to the handler source rather than to the model's own assertion.

Kadi, Crime Analytics Platform for the Karnataka State Police *(KSP Datathon 2026)* | **Live** | **GitHub** TypeScript, Node.js, Zoho Catalyst

- Built and deployed an end-to-end analytics platform on Zoho Catalyst over 50 days, joining 59,985 synthetic FIRs, generated on the real KSP schema, across 31 districts and 298 stations into one queryable link graph with per-FIR ego networks, 5 layouts and 6 link-type filters.
- Shipped a command dashboard with disposal funnels, hour-by-weekday heatmaps and forecasts with 95% bands, plus evidence OCR and a grounded bilingual English and Kannada assistant that refuses to translate FIR numbers, dates or identifiers.
- Built one ranked worklist over 26,168 flagged cases, ordering by severity then urgency against the peer median, each row carrying a plain-language reason and a next action.

CacheLLM, Semantic Caching Proxy for LLM APIs *(Published on PyPI)* | **Live** | **GitHub** Python, FastAPI, Embeddings, Redis, PyPI

- Teams pay full price to answer the same question reworded. Built a drop-in proxy that keeps the OpenAI request and response shape including streaming, verified against the official Python and Node SDKs and LangChain, so adopting it costs one base-URL change.
- Routed six providers behind one interface from each host's live model list, with the cache as an in-process numpy matrix by default and Redis optional, so it runs with nothing else installed.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL

Frontend: React, Next.js, TypeScript, Tailwind CSS, Vite, Recharts, MapLibre GL, Streamlit, Responsive UI

Backend and APIs: FastAPI, Pydantic, Node.js, Express, REST APIs, PostgreSQL, MongoDB, DuckDB, Redis, Zod Schema Validation

Infrastructure and Tooling: Docker, AWS, Google Cloud Run, Vercel, Render, GitHub Actions, CI/CD, Linux, Git, Observability, Reproducible Builds

Core Computer Science: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

ACHIEVEMENTS

- Adaptive Graph Search Suite: 12 routing algorithms in C++20 over 1.7M junctions, Contraction Hierarchies 44x faster than Dijkstra, verified exact against a reference implementation and compiled to WebAssembly. GitHub · Live
- Hacksplosion 2026, Deloitte India: cleared Levels 1, 2 and 3 with an AI-driven resume-to-job matching pipeline.
- 39 public repositories, most deployed and free to use, with every reported number reproducible from the repository and failures published beside results.

EDUCATION

The LNM Institute of Information Technology, Jaipur **2023 - 2027**
B.Tech, Computer Science and Engineering CGPA: 7.11 / 10

LEADERSHIP

- Coordinator, Debate Society, LNMIIT Jaipur (May 2025 - Apr 2026): led a 150+ member society across debates, discussions and inter-college competitions.
- Secretariat and Organizer, LNMIIT Model United Nations (Aug 2025 - Jan 2026): organised a large-scale MUN conference, coordinating delegates, committees, schedules and logistics.